Knowing-Value Logic with Successor Arithmetic

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Background and Motivation

Background

Epistemic Logic (EL), since its birth, has been mainly focusing on reasoning about propositional knowledge, the knowledge expressed by "i knows that ϕ". However, besides "knowing that", there are apparently various types of knowledge used in everyday life, expressed by "knowing what" (i knows what d is), "knowing how"(i knows how to do ϕ), "knowing whether"(i knows whether ϕ) and so on.

Knowing-Value Operators

Operator	Meaning	Logical Form	Example
$Kv_i c$	Agent i knows the value of the non-rigid constant c.	$\exists x\, K_i(c = x)$	"i knows what the password is."
$Kv_i(\phi, c)$	Agent i knows the value of c under condition ϕ.	$\exists x\, K_i(\phi \rightarrow c = x)$	"i knows the password, given that it is a 4-digit number."

Conditional Kv Logic: Language

Definition (Language $\mathbf{PALKv}^r$)

Given a countably infinite set of proposition letters $\mathbf{P}$, a countably infinite set of agent names $\mathbf{I}$, and a countably infinite set of (non-rigid) constant symbols $\mathbf{C}$, the language of $\mathbf{PALKv}^r$ is defined as follows:

$$\phi ::= \top \mid p \mid \neg\phi \mid \phi \wedge \phi \mid K_i\phi \mid Kv_i(\phi, c) \mid [\phi]\phi$$

where $p \in \mathbf{P}, i \in \mathbf{I}, c \in \mathbf{C}$.

We denote the announcement-free part of $\mathbf{PALKv}^r$ as $\mathbf{ELKv}^r$.

Conditional Kv Logic: Semantics

Definition (Models of $\mathbf{PALKv}^r$)

A model $\mathcal{M}$ is defined as $\langle W, \{\sim_i | \ i \in I\}, O, V, V_{\mathbf{C}} \rangle$, where W is a non-empty set of possible worlds, $\sim_i$ is an equivalence relation on W, O is a non-empty set of objects, V is a valuation function assigning a set of worlds to each $p \in \mathbf{P}$, and $V_{\mathbf{C}} : \mathbf{C} \times W \to O$ assigns an object for each constant-world pair.

Conditional Kv Logic: Semantics

The semantics is defined as follows:

$\mathcal{M}, w \models \top$		always holds
$\mathcal{M}, w \models p$	$\iff$	$w \in V(p)$
$\mathcal{M}, w \models \neg\phi$	$\iff$	$\mathcal{M}, w \not\models \phi$
$\mathcal{M}, w \models \phi \wedge \psi$	$\iff$	$\mathcal{M}, w \models \phi$ and $\mathcal{M}, w \models \psi$
$\mathcal{M}, w \models K_i\psi$	$\iff$	for all u such that $w \sim_i u$: $\mathcal{M}, u \models \psi$
$\mathcal{M}, w \models Kv_i(\phi, c)$	$\iff$	for any $u, v \in W$ such that $w \sim_i u$ and $w \sim_i v$: if $\mathcal{M}, u \models \phi$ and $\mathcal{M}, v \models \phi$, then $V_{\mathbf{C}}(c, u) = V_{\mathbf{C}}(c, v)$
$\mathcal{M}, w \models [\psi]\phi$	$\iff$	$\mathcal{M}, w \models \psi$ implies $\mathcal{M}\|_\psi, w \models \phi$

Limitations: Coarse-Grained Formalization

Sentence	Formalization in $\mathbf{ELKv}^r$
The password is a 4-digit number.	p
The password is 1111.	q
S knows the value of the password, given that it is a 4-digit number.	$Kv_S(p, c)$
S knows that if the password is 4-digit, then it is 1111.	$K_S(p \to q)$

In natural language, the second statement clearly implies the first. But in $\mathbf{ELKv}^r$, $K(p \to q)$ does not imply $Kv(p, c)$.

Limitations: No specific stuctures

a model for $\mathbf{ELKv}^r$ assigns values to constants over a bare, unstructured set of objects O. To study specific mathematical structures, such as natural numbers with the successor function $(\mathbb{N}, 0^{\mathbb{N}}, S^{\mathbb{N}})$ or with addition and multiplication $(\mathbb{N}, 0^{\mathbb{N}}, 1^{\mathbb{N}}, +^{\mathbb{N}}, \times^{\mathbb{N}})$, we must enrich the language with arithmetic function symbols and add arithmetic axioms to the proof system.

Nothing is rigid in $\mathbf{ELKv}^r$. However, when we reason about knowledge, we may refer to a specific fixed structure.

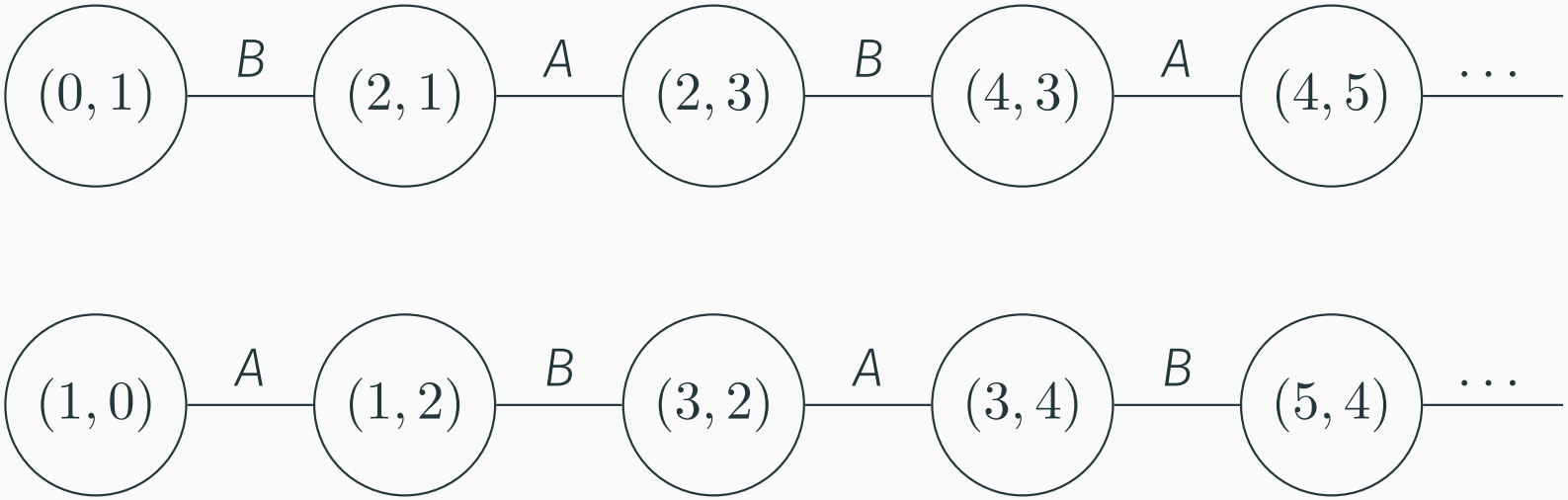
An Example: Consecutive Numbers

Anne and Bill get to hear the following: “Given are two natural numbers. They are consecutive numbers. I am going to whisper one of these numbers to Anne and the other number to Bill.” This happens. Anne and Bill now have the following conversation.

- Anne: “I don’t know your number.”
- Bill: “I don’t know your number.”
- Anne: “I know your number.”
- Bill: “I know your number.”

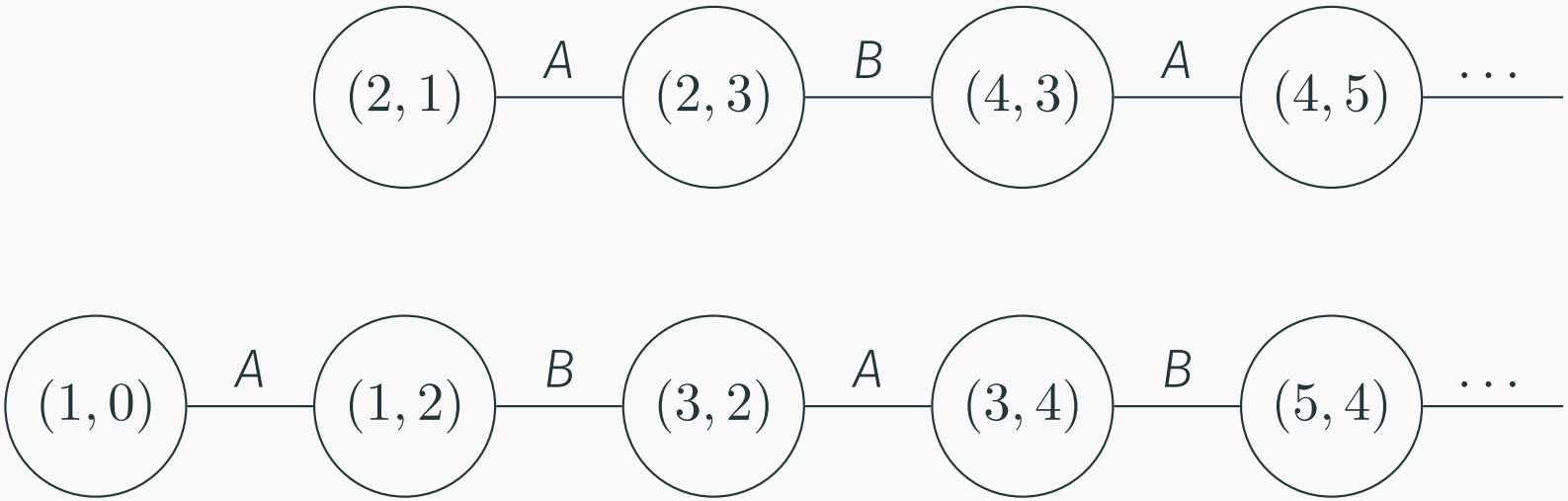
How is that possible? What surely is one of the two numbers?

Analysis



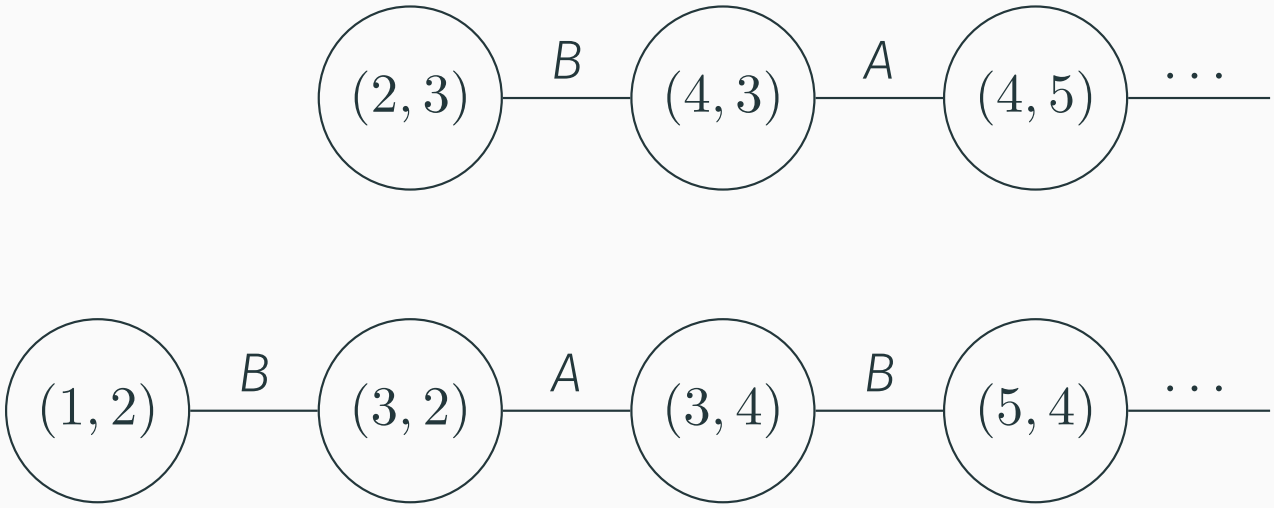
- Anne: “I don’t know your number.”

Analysis



- Anne: “I don’t know your number.”
- Bill: “I don’t know your number.”

Analysis



- Anne: “I don’t know your number.”
- Bill: “I don’t know your number.”
- Anne: “I know your number.”

Analysis





- Anne: “I don’t know your number.”
- Bill: “I don’t know your number.”
- Anne: “I know your number.”
- Bill: “I know your number.”

Formalization

To formalize the puzzle, the following statements are crucial:

- The two numbers are consecutive numbers.
- If Anna's number is 0, she knows Bill's number.

So it is natural to add some arithmetic to the logic. Here we add the successor function S, the equation symbol $\approx$, and the constant symbol 0. And for the semantics, we let the domain of the constant symbols be the set of natural numbers $\mathbb{N}$ with the usual successor function.

Knowing-Value Logic with Successor Arithmetic

Language

Definition (Language $\mathbf{PALKvSA}^r(\mathbf{I}, \mathbf{C})$)

Given a countable constant symbol set $\mathbf{C}$ and a countable set of agents $\mathbf{I}$, the set of terms Term and formulas are defined as follows.

$$t ::= 0 \mid c \mid St$$
$$\varphi ::= \top \mid t \approx t \mid \neg\varphi \mid \varphi \wedge \varphi \mid K_i\varphi \mid Kv_i(\varphi, t) \mid [\varphi]\varphi$$

where $c \in \mathbf{C}$, $i \in \mathbf{I}$, 0 is a rigid constant symbol, S is the successor function symbol. $t \approx t'$ is called an atomic formula and it replaces the role of propositional letters. We write $Kv_i t$ as an abbreviation of $Kv_i(\top, t)$.

We denote the announcement-free fragment of $\mathbf{PALKvSA}^r$ as $\mathbf{ELKvSA}^r$.

Semantics: Standard Models

Definition (Standard Models)

$\mathcal{M} = \langle W, \mathbb{N}, \mathbf{S}, \{\sim_i\}_{i \in \mathbf{I}}, V_{\mathbf{C}} \rangle$ is called a standard model, where $\sim_i$ is an equivalence relation on the set of possible worlds W for each agent i. Here $\mathbb{N}$ is the set of natural numbers and $\mathbf{S}$ is the successor function on $\mathbb{N}$, and $V_{\mathbf{C}} : \mathbf{C} \times W \to \mathbb{N}$ assigns a natural number for each constant-world pair.

Define the assignment for terms $V_{\mathrm{T}} : \mathrm{Term} \times W \to \mathbb{N}$ as follows:

$$V_{\mathrm{T}}(0, w) = \mathbf{0}$$
$$V_{\mathrm{T}}(c, w) = V_{\mathbf{C}}(c, w)$$
$$V_{\mathrm{T}}(St, w) = \mathbf{S}(V_{\mathrm{T}}(t, w))$$

Semantics

Then the satisfaction relation is defined as follows:

$$
\begin{array}{lcl}
\mathcal{M}, w \models t_1 \approx t_2 & \iff & V_{\mathrm{T}}(t_1, w) = V_{\mathrm{T}}(t_2, w) \\
\mathcal{M}, w \models Kv_i(\varphi, t) & \iff & \text{for any } u, v \in W \text{ such that } w \sim_i u, w \sim_i v, \\
 & & \text{if } \mathcal{M}, u \models \varphi \text{ and } \mathcal{M}, v \models \varphi, \text{ then} \\
 & & V_{\mathrm{T}}(t, u) = V_{\mathrm{T}}(t, v) \\
\mathcal{M}, w \models [\varphi]\psi & \iff & \mathcal{M}, w \models \varphi \text{ implies } \mathcal{M}|_\varphi, w \models \psi
\end{array}
$$

Proposition

$\mathbf{ELKvSA}^r$ *is not compact w.r.t. standard models.*

Proof.
Consider the formula set $\{\neg c \approx S^n 0 \mid n \in \mathbb{N}\}$. This type cannot be realized in standard model. □

Universal Formulas

For each arithmetic formula $\alpha(c_1, \cdots, c_n)$,

$$\models \alpha(c_1, \cdots, c_n) \iff \mathfrak{N} \models \forall x_1, \cdots, x_n \alpha(x_1, \ldots, x_n)$$

Semantics: Non-standard Models

Definition (Non-standard Models)

$(\mathbb{N}^\star, \mathbf{S}^\star)$ is a ω-saturated non-standard model of first order successor arithmetic, where $\mathbb{N}^\star = \mathbb{N} \cup \{(n, z) \mid n \in \mathbb{N}, z \in \mathbb{Z}\}$, and $\mathbf{S}^\star(n) = n + 1$, $\mathbf{S}^\star(n, z) = (n, z + 1)$. So we define the non-standard model for our logic as follows:

$$\mathcal{N} = \langle W, \mathbb{N}^\star, \mathbf{S}^\star, \{\sim_i\}_{i \in I}, V^\star_{\mathbf{C}} \rangle$$

The definition of $V^\star_{\mathbf{C}}$, $V^\star_{\mathrm{T}}$ and the satisfaction relation are the same as before, just replacing $(\mathbb{N}, \mathbf{S})$ with $(\mathbb{N}^\star, \mathbf{S}^\star)$. For convenience, let $Z_n := \{(n, z) \mid z \in \mathbb{Z}\}$.

Preparations for Axiomatization

By reduction, $\mathbb{ELKV}^r$ plus !ATOM, !NEG, !CON, !K, !Kv^r and RE or !COM completely axiomatize $\mathbf{PALKv}^r$.

!ATOM	$[\psi]p \leftrightarrow (\psi \to p)$
!NEG	$[\psi]\neg\phi \leftrightarrow (\psi \to \neg[\psi]\phi)$
!CON	$[\psi](\phi \wedge \chi) \leftrightarrow ([\psi]\phi \wedge [\psi]\chi)$
!K	$[\psi]K_i\phi \leftrightarrow (\psi \to K_i[\psi]\phi)$
!Kv^r	$[\phi]Kv_i(\psi, c) \leftrightarrow \big(\phi \to Kv_i\big(\phi \wedge [\phi]\psi, c\big)\big)$
!COM	$[\psi][\chi]\phi \leftrightarrow [\psi \wedge [\psi]\chi]\phi$

Preparations for Axiomatization

System $\mathbb{ELKV}^r$

Axiom Schemas	
TAUT	all the instances of tautologies
DISTK	$K_i(\phi \to \psi) \to (K_i\phi \to K_i\psi)$
T	$K_i\phi \to \phi$
4	$K_i\phi \to K_iK_i\phi$
5	$\neg K_i\phi \to K_i\neg K_i\phi$
DISTKv^r	$K_i(\phi \to \psi) \to (Kv_i(\psi, c) \to Kv_i(\phi, c))$
Kv^{r}4	$Kv_i(\phi, c) \to K_iKv_i(\phi, c)$
Kv$^r\bot$	$Kv_i(\bot, c)$
Kv$^r\vee$	$\widehat{K}_i(\phi \wedge \psi) \wedge Kv_i(\phi, c) \wedge Kv_i(\psi, c) \to Kv_i(\phi \vee \psi, c)$

Preparations for Axiomatization

Rules

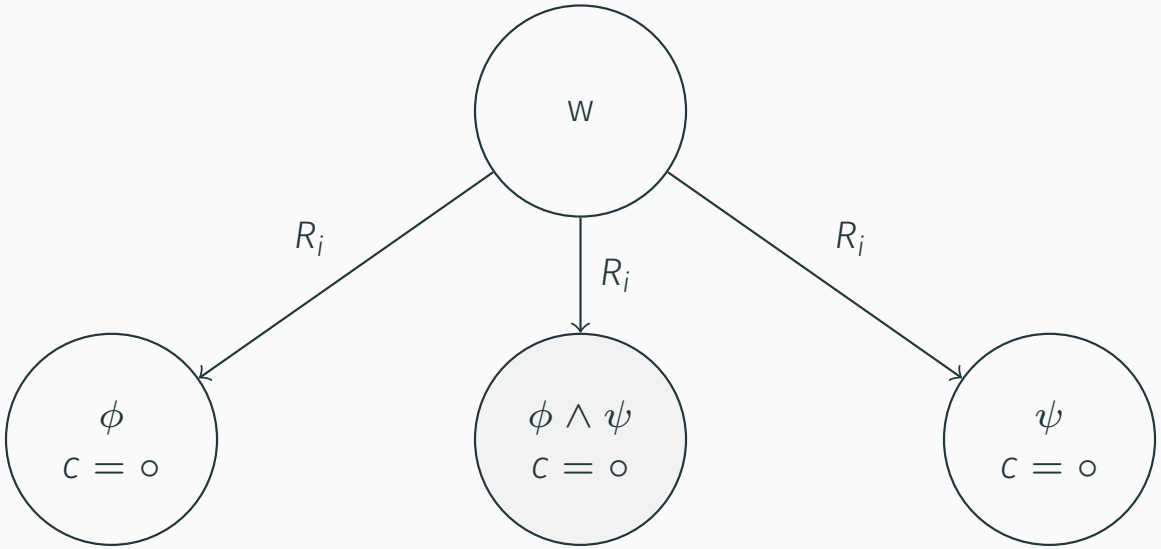
MP $\dfrac{\phi, \phi \to \psi}{\psi}$

NECK $\dfrac{\phi}{K_i \phi}$

RE $\dfrac{\psi \leftrightarrow \chi}{\phi \leftrightarrow \phi[\psi/\chi]}$

Understanding $\mathsf{Kv}^r\vee$

$$\widehat{K}_i(\phi \wedge \psi) \wedge Kv_i(\phi, c) \wedge Kv_i(\psi, c) \rightarrow Kv_i(\phi \vee \psi, c)$$



- $\widehat{K}_i(\phi \wedge \psi)$ gives an accessible overlap world.
- $Kv_i(\phi, c)$ forces the same value on the accessible ϕ-worlds.
- $Kv_i(\psi, c)$ forces the same value on the accessible ψ-worlds.

Thus all accessible $\phi \vee \psi$ worlds have the same value of c.

Successor Arithmetic

Theorem

Given the symbol set $\{0, S\}$ and the structure $\mathfrak{N} = (\mathbb{N}, \mathbf{S}, \mathbf{0})$ where $\mathbb{N}$ is the set of natural numbers and $\mathbf{S}$ is the successor function on it, the first order theory of this structure $Th(\mathfrak{N})$ is axiomatized by the following three axioms and an additional axiom schema.

$$\forall x(\neg Sx \approx 0)$$

$$\forall x \forall y(Sx \approx Sy \to x \approx y)$$

$$\forall x(\neg x \approx 0 \to \exists y x \approx Sy)$$

$\forall x(\neg S^n(x) \approx x)$ *for every natural number* $n \geq 1$

where $S^n(x)$ *is an abbreviation of* $\underbrace{SS \cdots S}_{n \text{ times}}(x)$.

Axiom System $\mathbb{ELKVSA}^r$

Axiom Schemas

TAUT	all the instances of tautologies
DISTK	$K_i(\varphi \to \psi) \to (K_i\varphi \to K_i\psi)$
T	$K_i\varphi \to \varphi$
4	$K_i\varphi \to K_iK_i\varphi$
5	$\neg K_i\varphi \to K_i\neg K_i\varphi$
ID	$t \approx t$
SI	$t \approx t' \to (\alpha \to \alpha(t'//t))$
	where α is an atomic formula
SA1	$\neg 0 \approx St$
SA2	$St_1 \approx St_2 \to t_1 \approx t_2$
SA3	$\neg S^n t \approx t$, where $n \geq 1$
Kv^r0	Kv_i0
$\mathrm{Kv}^r\mathrm{S}$	$Kv_i(\varphi, t) \leftrightarrow Kv_i(\varphi, St)$
Kv^r4	$Kv_i(\varphi, t) \to K_iKv_i(\varphi, t)$
KVE1	$Kv_it_1 \wedge K_i(\varphi \to t_1 \approx t_2) \to Kv_i(\varphi, t_2)$
KVE2	$Kv_it_1 \wedge Kv_it_2 \to (K_it_1 \approx t_2 \vee K_i\neg t_1 \approx t_2)$

Axiom System $\mathbb{ELKVSA}^r$

Rules

MP
$$\frac{\varphi \quad \varphi \to \psi}{\psi}$$

NECK
$$\frac{\varphi}{K_i\varphi}$$

RE
$$\frac{\psi \leftrightarrow \chi}{\varphi \leftrightarrow \varphi[\psi/\chi]}$$

Kv^r-Elim
$$\frac{(Kv_i c \wedge K_i(\varphi \to c \approx t)) \to \psi}{Kv_i(\varphi, t) \to \psi}$$

where c does not occur in φ, ψ, t

Intuition for the Axioms

Axiom		Intuition
$\mathrm{Kv}^r 0$	$Kv_i 0$	0 is a rigid constant.
$\mathrm{Kv}^r \mathrm{S}$	$Kv_i(\varphi, t) \leftrightarrow Kv_i(\varphi, St)$	S is a rigid injection.

The axiom $\mathrm{Kv}^r \mathrm{S}$ plays a crucial role in our completeness proof, enabling us to reduce the general case of an arbitrary term t to that of a constant c.

Intuition for the Axioms

KVE1 $\quad Kv_i t_1 \wedge K_i(\varphi \to t_1 \approx t_2) \to Kv_i(\varphi, t_2)$

KVE2 $\quad Kv_i t_1 \wedge Kv_i t_2 \to (K_i t_1 \approx t_2 \vee K_i \neg t_1 \approx t_2)$

Kv^r-Elim $\quad \dfrac{(Kv_i c \wedge K_i(\varphi \to c \approx t)) \to \psi}{Kv_i(\varphi, t) \to \psi}$

where c does not occur in φ, ψ, t

Mention-some reading		Mention-all reading	
$\exists x\, K_i(c \approx x)$		$\forall x\, (K_i(c \approx x) \vee K_i \neg(c \approx x))$	
KVE1	$\exists$-Intro	KVE2	$\exists$-Intro/$\forall$-Elim
Kv^r-Elim	$\exists$-Elim	?	$\exists$-Elim

$\neg \mathsf{Kv}^r$-Elim is hidden

$$\frac{\big(\mathsf{Kv}_i c \wedge \widehat{K}_i(\varphi \wedge c \approx t) \wedge \widehat{K}_i(\varphi \wedge \neg c \approx t)\big) \to \psi}{\neg \mathsf{Kv}_i(\varphi, t) \to \psi} \quad (c \text{ fresh}).$$

Examples of the role it can play:

$$\mathsf{St}_1 \approx \mathsf{St}_2 \to t_1 \approx t_2 \quad \rightsquigarrow \quad \mathsf{Kv}_i(\mathsf{St}) \to \mathsf{Kv}_i t,$$

$$t_1 + t_2 = t_1' + t_2' \wedge t_1 = t_1' \to t_2 = t_2' \quad \rightsquigarrow \quad \mathsf{Kv}_i(t_1 + t_2) \wedge \mathsf{Kv}_i t_1 \to \mathsf{Kv}_i t_2.$$

Power of Kv^r-Elim

Proposition
The following theorems can be derived in $\mathbb{ELKVSA}^r$*:*

1. $\mathit{DISTKv}^r : K_i(\varphi \to \psi) \to (\mathsf{Kv}_i(\psi, t) \to \mathsf{Kv}_i(\varphi, t))$
2. $\mathsf{Kv}^r\bot : \mathsf{Kv}_i(\bot, t)$
3. $\mathsf{Kv}^r\vee : \widehat{K}_i(\phi \wedge \psi) \wedge \mathsf{Kv}_i(\phi, t) \wedge \mathsf{Kv}_i(\psi, t) \to \mathsf{Kv}_i(\phi \vee \psi, t)$

Completeness Proof

Completeness Proof Strategy

- We first prove the strong completeness of $\mathbb{ELKVSA}^r$ w.r.t. non-standard models.
- Then we show that if ϕ is satisfiable in a non-standard model, it is satisfiable in a finite non-standard model.
- Finally, we show that if ϕ is satisfiable in a finite non-standard model, it is satisfiable in a finite standard model.
- Thus, We can conclude that $\mathbb{ELKVSA}^r$ is weakly complete w.r.t. standard models.

Completeness

Wang and Fan (2013) proved the completeness of $\mathbb{ELKV}^r$ for the single agent case. The multi-agent case was later solved by them in 2014, employing a profound and insightful, but highly complex technique known as “fake formulas”.

In our work, we claim that the only steps we need are:

- Tree-Unraveling
- Play the Valuation Game

Cononical Model

We first consider the classical canonical frame $\mathcal{F}^c = \langle W^c, \{\sim_i^c\}_{i \in I} \rangle$, where

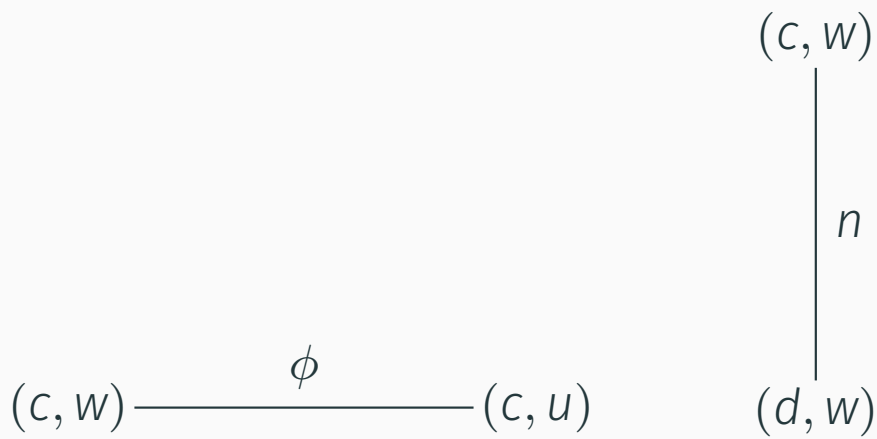
- $W^c = MCS \times \{1, 2\}$, where MCS is the set of all maximal consistent sets.
- For any $w, v \in W^c$, $w \sim_i^c v$ iff for any formula φ, $K_i\varphi \in w \Rightarrow \varphi \in v$. [1]

Obviously, $\sim_i^c$ is an equivalence relation on W^c.

[1] Strictly speaking, $\phi \in w$ means $\phi \in \pi_1(w)$, where π_1 is the projection function from W^c to MCS. We omit π_1 for simplicity.

Valuation Game for Single-Agent Case

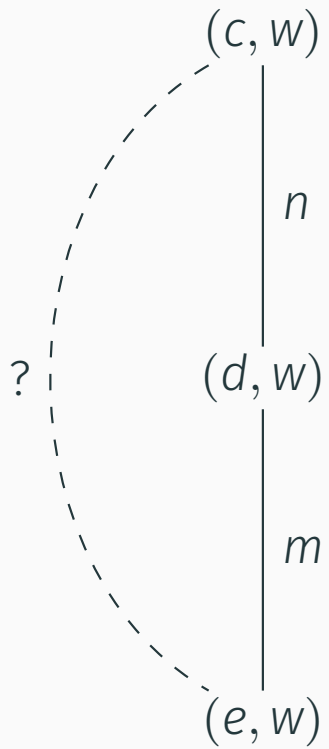
Consider constant-world pairs. The question is: when two pairs (c, w) and (d, u)[2] are forced to be in the same component of $\mathbb{N}^\star$? There are two basic cases:



$Kv_i(\phi, c) \wedge \phi \in w \cap u$ $\qquad$ $d \approx S^n c \in w$

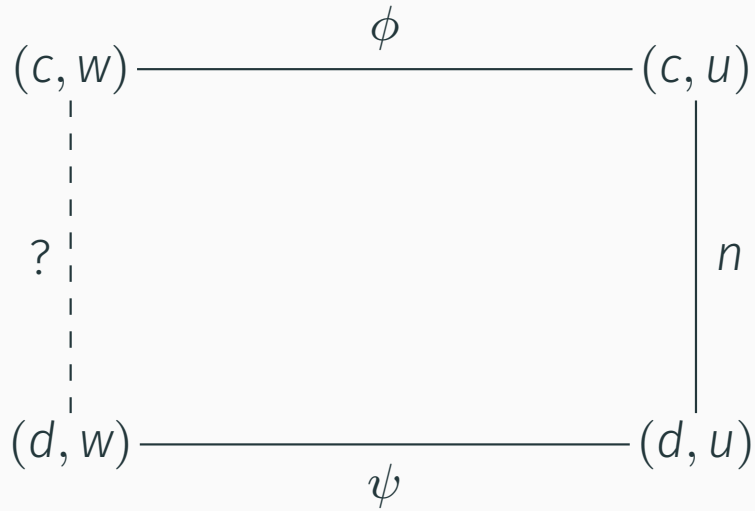
[2] c and d can also represent 0.

Where We Need Arithmetic



Arithmetic Axioms can help.

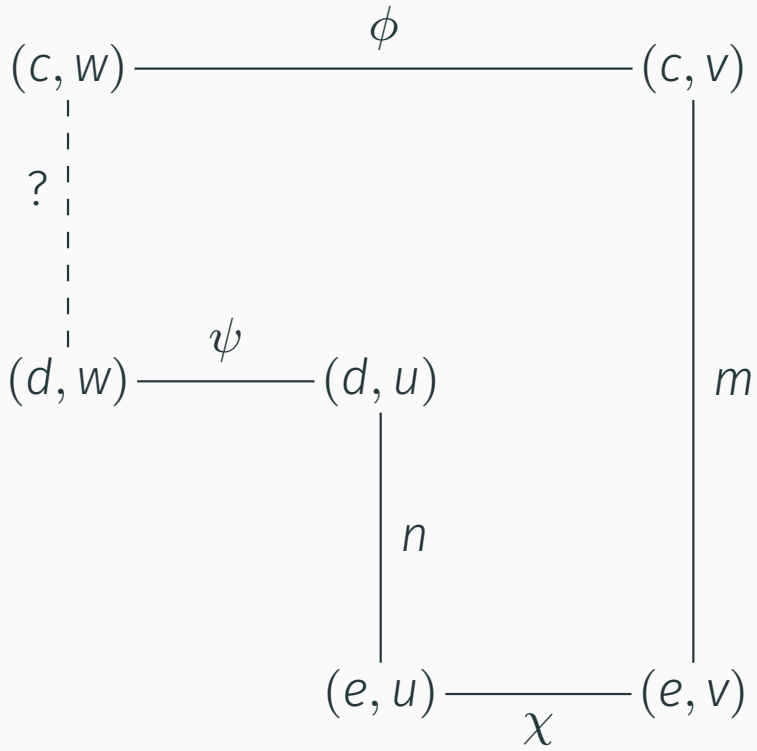
Combining two basic cases



We need an axiom:

$Kv_i(\phi, c) \wedge Kv_i(\psi, d) \rightarrow (\widehat{K}_i(\phi \wedge \psi \wedge d \approx S^n c) \rightarrow K_i(\phi \wedge \psi \rightarrow d \approx S^n c))$

Combining two basic cases



We need an axiom: $Kv_i(\phi, c) \wedge Kv_i(\psi, d) \wedge Kv_i(\chi, e) \rightarrow (\widehat{K}_i(\psi \wedge \chi \wedge e \approx S^n d) \wedge \widehat{K}_i(\phi \wedge \chi \wedge e \approx S^m c) \rightarrow K_i(\phi \wedge \psi \rightarrow d \approx S^{m-n} c))$.

Combining two basic cases

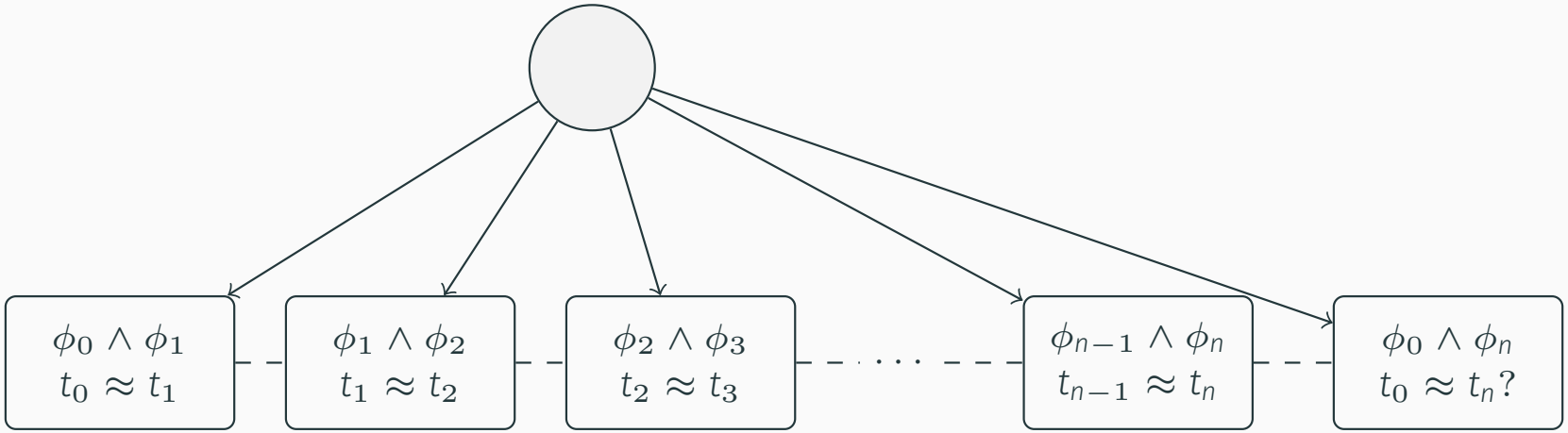
In general, we need the following axiom schema:

Proposition

GKVE is derivable in the system $\mathbb{ELKVSA}^r$.

- *GKVE:* $\left(\bigwedge_{k=0}^{n} Kv_i(\phi_k, t_k)\right) \wedge \left(\bigwedge_{k=0}^{n-1} \widehat{K}_i(\phi_k \wedge \phi_{k+1} \wedge t_k \approx t_{k+1})\right) \rightarrow K_i\,(\phi_0 \wedge \phi_n \rightarrow t_0 \approx t_n)$

Why GKVE is Valid



Deriving GKVE

We introduce $n+1$ fresh, pairwise distinct constants $c_0, \ldots, c_n$ which do not occur in ϕ_k, t_k.

$$\Gamma := \bigwedge_{k=0}^{n} \left(Kv_i c_k \wedge K_i(\phi_k \to c_k \approx t_k) \right)$$

$$\Delta := \bigwedge_{k=0}^{n-1} \widehat{K}_i(\phi_k \wedge \phi_{k+1} \wedge t_k \approx t_{k+1}).$$

$$\psi := K_i \left(\phi_0 \wedge \phi_n \to t_0 \approx t_n \right).$$

We only need to prove that $\Gamma \to (\Delta \to \psi)$.

Deriving GKVE

Assume Γ and Δ.

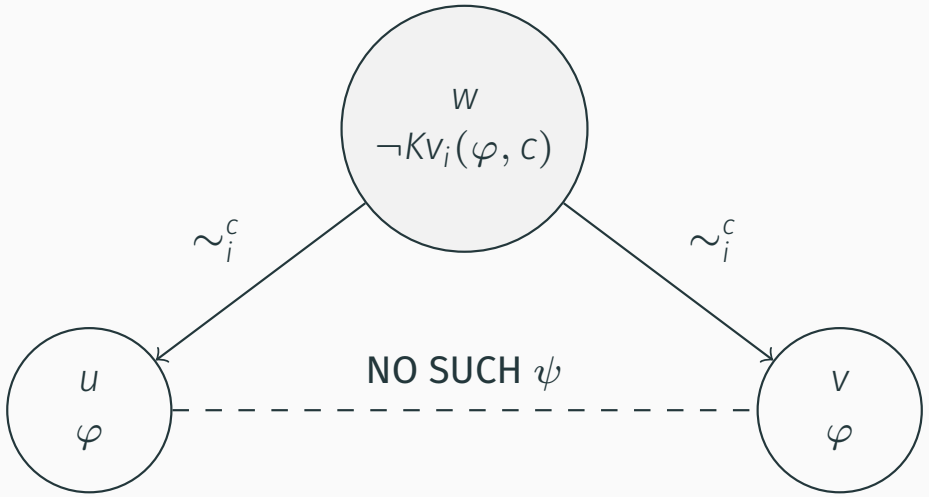
First, for each $0 \leq k \leq n-1$, from $K_i(\phi_k \to c_k \approx t_k)$ and $K_i(\phi_{k+1} \to c_{k+1} \approx t_{k+1})$, we can derive $K_i\big((\phi_k \wedge \phi_{k+1} \wedge t_k \approx t_{k+1}) \to c_k \approx c_{k+1}\big)$. Combining this with $\widehat{K}_i(\phi_k \wedge \phi_{k+1} \wedge t_k \approx t_{k+1})$ from Δ yields $\widehat{K}_i(c_k \approx c_{k+1})$. By KVE2 and $Kv_i c_k$ and $Kv_i c_{k+1}$, we derive $K_i(c_k \approx c_{k+1})$ for each $0 \leq k \leq n-1$. By the transitivity of equality inside the K_i modality, this naturally yields $K_i(c_0 \approx c_k)$ for all $0 \leq k \leq n$.

- For GKVE: With $K_i(c_0 \approx c_n)$ derived, along with $K_i(\phi_0 \to c_0 \approx t_0)$ and $K_i(\phi_n \to c_n \approx t_n)$ from Γ, standard equality reasoning yields $K_i(\phi_0 \wedge \phi_n \to t_0 \approx t_n)$.

Witnesses for $\neg Kv_i(\varphi, c)$

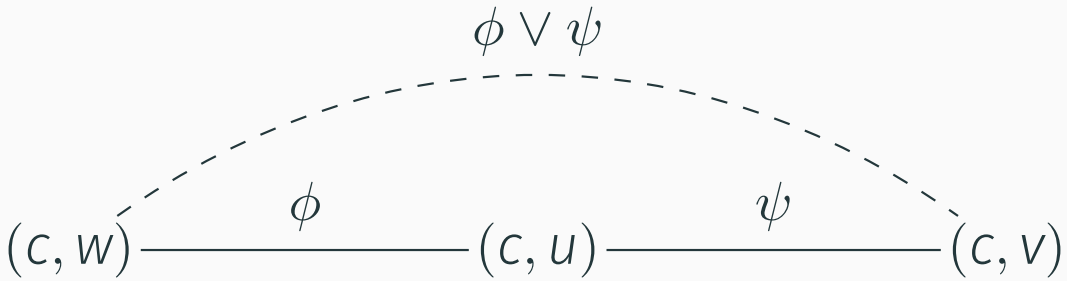
Lemma

Suppose $\Psi_c^i(w) = \{\psi \mid Kv_i(\psi, c) \in w\}$. *If* $\neg Kv_i(\varphi, c) \in w$, *then we can construct two different worlds* $u, v \in W^c$ *such that* $w \sim_i^c u$, $w \sim_i^c v$, $\varphi \in u \cap v$, *but* $u \cap v \cap \Psi_c^i(w) = \emptyset$.



$u \cap v \cap \Psi_c^i(w) = \emptyset$
no ψ with $Kv_i(\psi, c) \in w$ is shared by both witnesses

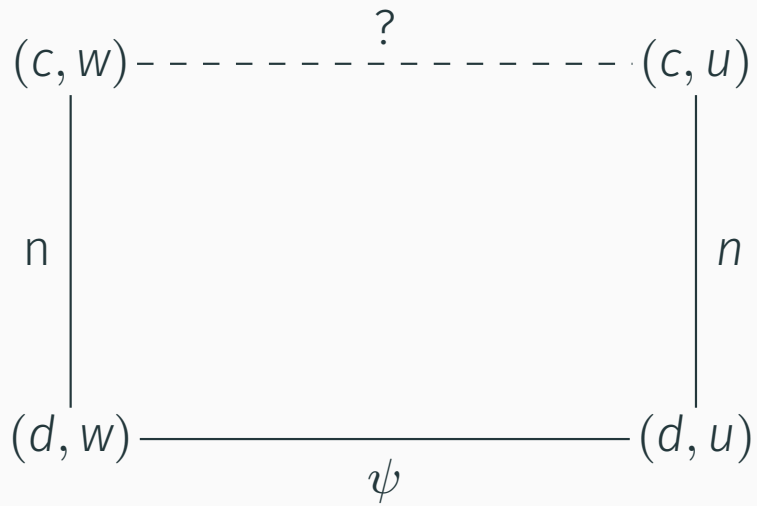
Witnesses for $\neg Kv_i(\varphi, c)$



$$(c, w) \xrightarrow{\phi} (c, u) \xrightarrow{\psi} (c, v) \quad \Longrightarrow \quad (c, w) \xrightarrow{\phi \vee \psi} (c, v)$$

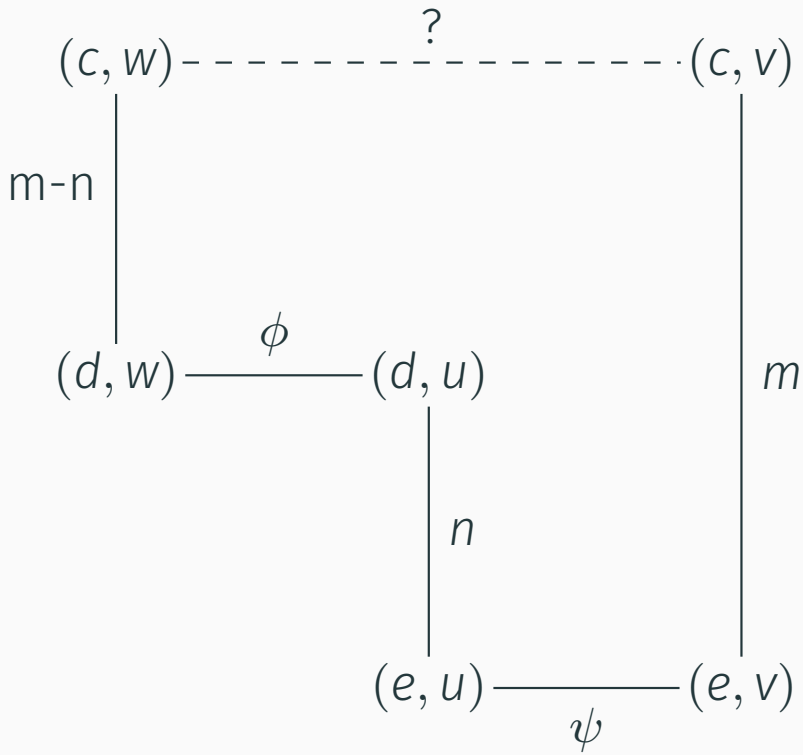
We need the axiom:
$\mathrm{Kv}^r\vee : \widehat{K}_i(\phi \wedge \psi) \wedge Kv_i(\phi, c) \wedge Kv_i(\psi, c) \to Kv_i(\phi \vee \psi, c)$.

Witnesses for $\neg Kv_i(\varphi, c)$



We need an axiom: $Kv_i(\psi, d) \to Kv_i(\psi \wedge d \approx S^n c, c)$.

Witnesses for $\neg Kv_i(\varphi, c)$



We need an axiom: $Kv_i(\phi, d) \wedge Kv_i(\psi, e) \wedge \widehat{K}_i(\psi \wedge \phi \wedge e \approx S^n d) \rightarrow Kv_i(\phi \wedge d \approx S^{m-n} c) \vee (\psi \wedge e \approx S^m c), c)$.

Witnesses for $\neg Kv_i(\varphi, c)$

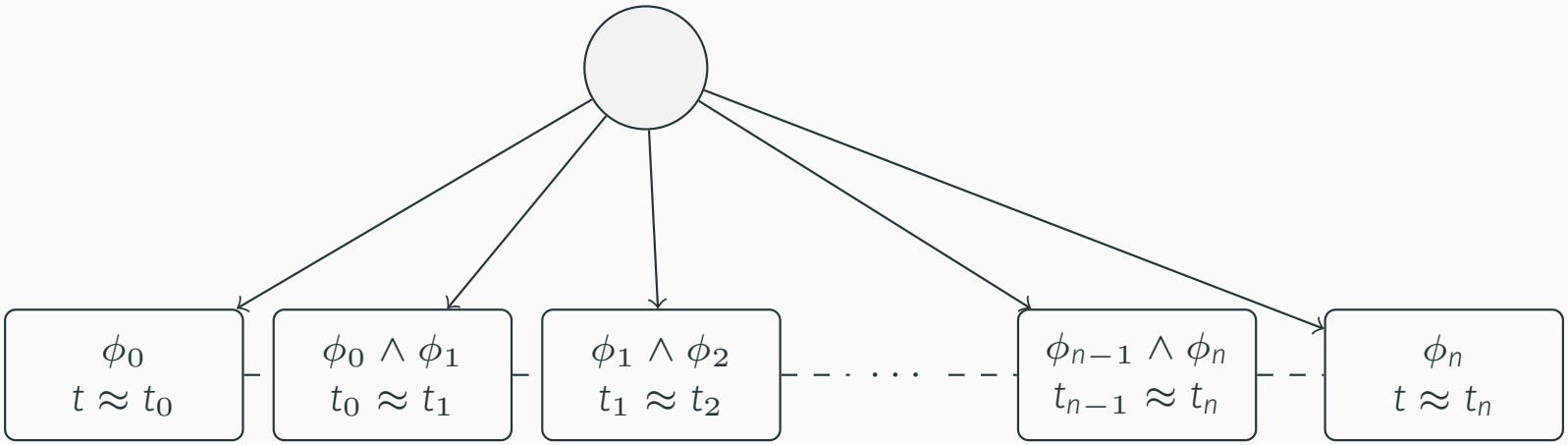
In general, we need the following proposition:

Proposition

$GKv^r\vee$ are derivable in the system $\mathbb{ELKVSA}^r$.

- $GKv^r\vee$: $\left(\bigwedge_{k=0}^{n} Kv_i(\phi_k, t_k)\right) \wedge \left(\bigwedge_{k=0}^{n-1} \widehat{K}_i(\phi_k \wedge \phi_{k+1} \wedge t_k \approx t_{k+1})\right) \rightarrow Kv_i\left(\bigvee_{k=0}^{n}(\phi_k \wedge t \approx t_k), t\right)$

Why GKv$^r\vee$ is Valid



Deriving GKv$^r\vee$

We introduce $n+1$ fresh, pairwise distinct constants $c_0, \ldots, c_n$ which do not occur in ϕ_k, t_k or t.

$$\Gamma := \bigwedge_{k=0}^{n} \left(Kv_i c_k \wedge K_i(\phi_k \to c_k \approx t_k) \right)$$

$$\Delta := \bigwedge_{k=0}^{n-1} \widehat{K}_i(\phi_k \wedge \phi_{k+1} \wedge t_k \approx t_{k+1}).$$

$$\psi := Kv_i \left(\bigvee_{k=0}^{n} (\phi_k \wedge t \approx t_k), t \right).$$

We only need to prove that $\Gamma \to (\Delta \to \psi)$.

Deriving GKv$^r\vee$

Assume Γ and Δ.

First, for each $0 \leq k \leq n-1$, from $K_i(\phi_k \to c_k \approx t_k)$ and $K_i(\phi_{k+1} \to c_{k+1} \approx t_{k+1})$, we can derive $K_i\big((\phi_k \wedge \phi_{k+1} \wedge t_k \approx t_{k+1}) \to c_k \approx c_{k+1}\big)$. Combining this with $\widehat{K}_i(\phi_k \wedge \phi_{k+1} \wedge t_k \approx t_{k+1})$ from Δ yields $\widehat{K}_i(c_k \approx c_{k+1})$. By KVE2 and $Kv_i c_k$ and $Kv_i c_{k+1}$, we derive $K_i(c_k \approx c_{k+1})$ for each $0 \leq k \leq n-1$. By the transitivity of equality inside the K_i modality, this naturally yields $K_i(c_0 \approx c_k)$ for all $0 \leq k \leq n$.

- For GKv$^r\vee$: From $K_i(\phi_k \to c_k \approx t_k)$ in Γ and $K_i(c_0 \approx c_k)$ derived above, we deduce $K_i(\phi_k \to c_0 \approx t_k)$, which implies $K_i\big((\phi_k \wedge t \approx t_k) \to c_0 \approx t\big)$. Taking the disjunction of the antecedents gives: $K_i\left(\bigvee_{k=0}^{n}(\phi_k \wedge t \approx t_k) \to c_0 \approx t\right)$. Since we have $Kv_i c_0$ from Γ, by KVE1, we can deduce $Kv_i\left(\bigvee_{k=0}^{n}(\phi_k \wedge t \approx t_k), t\right)$.

Enjoy the Valuation Game

Do the following:

- Select countable many worlds from the canonical model.
- If there exists a finite path from (c, w) to (d, u), they should be in the same component of $\mathbb{N}^\star$. If not, they should be in different components.
- If there exists two different paths from (c, w) to (d, u), the weight of the paths should be the same.
- Verify the following are satisfied: $t_1 \approx t_2$, $\neg t_1 \approx t_2$, $Kv(\phi, t)$, $\neg Kv(\phi, t)$.

The Multi-Agent Case

We give an example:

$$\{Kv_1c, \neg Kv_1d, Kv_2d, \neg Kv_2c, c \approx d\}$$

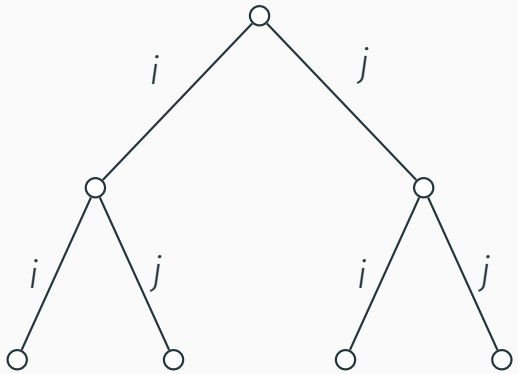
$$1 \mid 2$$

$$\{Kv_1c, \neg Kv_1d, Kv_2d, \neg Kv_2c, \neg c \approx d\}$$

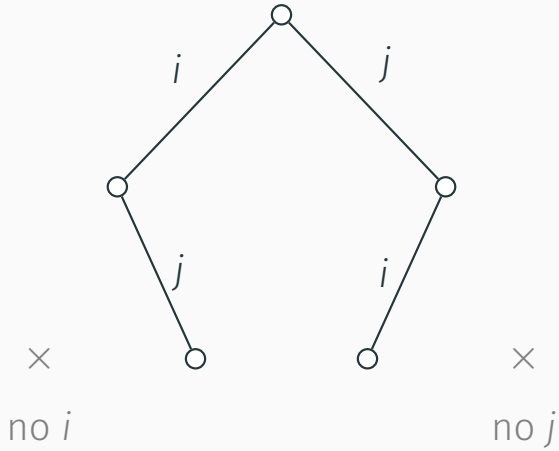
We cannot give a valuation! So tree-unraveling is needed.

Tree-Unraveling

Direct unraveling



Our tree-like frame



After S5 closure, each $\sim_i'$-class is one parent node plus its immediate i-successors. For $i \neq j$, an i-equvalence class and a j-equivalence class can only intersect at one node h.

Then there exist an finite path from (c, h) to (d, h) in the i-class iff there exist an finite path from (c, h) to (d, h) in the j-class.

Completeness w.r.t. Non-standard Models

Theorem

$\mathbb{ELKVSA}^r$ *is strongly complete w.r.t. non-standard models.*

Finite Model Property

Theorem

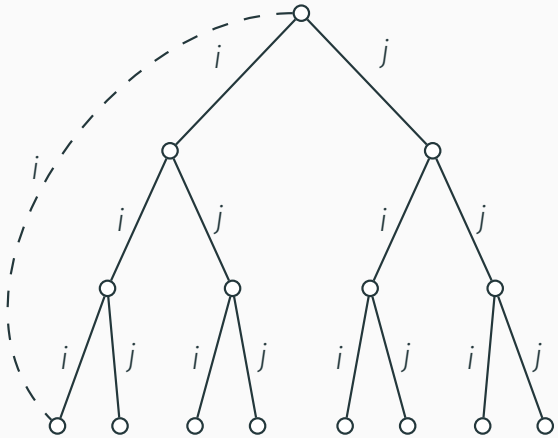
$\mathbf{ELKvSA}^r$ *has finite model property w.r.t. non-standard models. Moreover, if a formula* ϕ *is satisfiable in a non-standard model, it is satisfiable in a finite non-standard model whose size is bounded by* $(2|\Sigma|)^{\mathrm{depth}(\phi)+1}$*, where* Σ *is the set of subformulas of* ϕ *closed under single negation, and* $\mathrm{depth}(\phi)$ *is the modal depth of* ϕ*.*

- $\mathrm{depth}(\alpha) = 0$ for atomic formulas and Boolean combinations of atomics.
- $\mathrm{depth}(\neg\varphi) = \mathrm{depth}(\varphi)$, $\mathrm{depth}(\varphi \wedge \psi) = \max(\mathrm{depth}(\varphi), \mathrm{depth}(\psi))$.
- $\mathrm{depth}(K_i\varphi) = \mathrm{depth}(\varphi) + 1$.
- $\mathrm{depth}(Kv_i(\varphi, t)) = \mathrm{depth}(\varphi) + 1$.

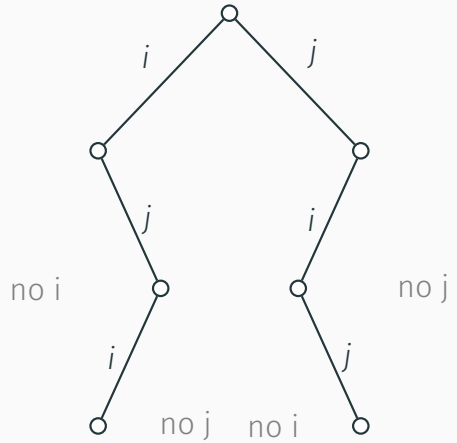
Let $D = \mathrm{depth}(\phi)$.

Selection

Direct selection



Our selection



If we select nodes directly, the hierarchy of depth will collapse after taking the S5 closure.

After our selection, we can prove the FMP by induction on the depth of formula.

Finite Model Property

For any formula ϕ satisfiable in a non-standard model, it is satisfiable in the tree-like model $\mathcal{M} = \langle W', \mathbb{N}^\star, \mathbf{S}^\star, \{\sim_i'\}_{i\in I}, V^\star_{\mathbf{C}}\rangle$.

Construct $W_s \subset W'$ level by level. Let $H'_0 = H_0 = \{(w_0)\}$. Assume H'_k is constructed for $k < D$. We construct H'_{k+1} as follows. For each $\vec{h} \in H'_k$ and $j \in I$:

1. If $j = \mathrm{Ag}(\vec{h})$: Skip.
2. If $j \neq \mathrm{Ag}(\vec{h})$:
   - For $\neg K_j\psi \in \Sigma$: If $\mathcal{M}, \vec{h} \models \neg K_j\psi$ and $\mathrm{depth}(\neg K_j\psi) \leq D - k$, select one witness $\vec{h'} \in H_{k+1}$ with $\mathcal{M}, \vec{h'} \not\models \psi$ and add to H'_{k+1}.
   - For $\neg Kv_j(\chi, S^k c) \in \Sigma$: If $\mathcal{M}, \vec{h} \models \neg Kv_j(\chi, S^k c)$ and $\mathrm{depth}(\neg Kv_j(\chi, S^k c)) \leq D - k$, select two witnesses $\vec{h}_1, \vec{h}_2 \in H_{k+1}$ with $\mathcal{M}, \vec{h}_1 \models \chi$ and $\mathcal{M}, \vec{h}_2 \models \chi$ such that $V^\star_{\mathbf{C}}(c, \vec{h}_1) \neq V^\star_{\mathbf{C}}(c, \vec{h}_2)$. Add them to H'_{k+1}. (This can be done because of the construction of W'.)

Let $W_s = \bigcup_{k=0}^{D} H'_k$. W_s is finite. Let $\sim_i^s$ be the restriction of $\sim_i'$ to W_s, and $V^s_{\mathbf{C}}$ be the restriction of $V^\star_{\mathbf{C}}$ to W_s. Let $\mathcal{M}_s = \langle W_s, \mathbb{N}^\star, \mathbf{S}^\star, \{\sim_i^s\}_{i\in I}, V^s_{\mathbf{C}}\rangle$.

Finite Model Property

By induction on the **modal depth** d ($0 \leq d \leq D$), we show that for each d such that $0 \leq d \leq D$ and for each formula $\psi \in \Sigma$ with $\mathrm{depth}(\psi) = d$, and for any node $\vec{h} \in W_s$, if the height condition $\mathrm{height}(\vec{h}) \leq D - d$ is satisfied, then: $\mathcal{M}_s, \vec{h} \models \psi \iff \mathcal{M}, \vec{h} \models \psi$.

Base Case ($d = 0$): ψ is atomic or Boolean combination of atomics. The condition $\mathrm{height}(\vec{h}) \leq D$ always holds for $\vec{h} \in W_s$. Since valuations are inherited directly from $\mathcal{M}$, the equivalence holds trivially.

Inductive Hypothesis: Assume the Claim holds for all formulas with depth $< d$, i.e., for any formula $\chi \in \Sigma$ with $\mathrm{depth}(\chi) < d$, and for any node $\vec{h} \in W_s$ satisfying $\mathrm{height}(\vec{h}) \leq D - \mathrm{depth}(\chi)$, we have: $\mathcal{M}_s, \vec{h} \models \chi \iff \mathcal{M}, \vec{h} \models \chi$.

Finite Model Property

Subcase 2: $\psi := \neg Kv_j(\chi, S^k c)$. Here $\mathrm{depth}(\psi) = d$, so $\mathrm{depth}(\chi) = d - 1$. Assume $\mathrm{height}(\vec{h}) \leq D - d$. ($\Leftarrow$) Suppose $\mathcal{M}, \vec{h} \models \neg Kv_j(\chi, S^k c)$.

- If $j \neq \mathrm{Ag}(\vec{h})$: Since $\mathrm{height}(\vec{h}) \leq D - d$, the selection condition is satisfied, so there exist two selected witnesses $\vec{h}_1, \vec{h}_2 \in H'_{\mathrm{height}(\vec{h})+1}$ with $\vec{h} \sim'_j \vec{h}_1$, $\vec{h} \sim'_j \vec{h}_2$, $\mathcal{M}, \vec{h}_1 \models \chi$, $\mathcal{M}, \vec{h}_2 \models \chi$, and $V^\star_{\mathbf{C}}(c, \vec{h}_1) \neq V^\star_{\mathbf{C}}(c, \vec{h}_2)$. By the IH (applicable since $\mathrm{height}(\vec{h}_\ell) \leq D - \mathrm{depth}(\chi), \ell = 1, 2$), $\mathcal{M}_s, \vec{h}_1 \models \chi$ and $\mathcal{M}_s, \vec{h}_2 \models \chi$. Since $\vec{h} \sim^s_j \vec{h}_1 \sim^s_j \vec{h}_2$, we get $\mathcal{M}_s, \vec{h} \models \neg Kv_j(\chi, S^k c)$.
- If $j = \mathrm{Ag}(\vec{h})$: Let $\vec{p}$ be the parent of $\vec{h}$. Then $\mathcal{M}, \vec{p} \models \neg Kv_j(\chi, S^k c)$ and $\mathrm{height}(\vec{p}) = \mathrm{height}(\vec{h}) - 1 < D - d$, so the selection condition for $\vec{p}$ is satisfied. Hence there exist two selected witnesses $\vec{w}_1, \vec{w}_2 \in H'_{\mathrm{height}(\vec{p})+1}$ in the same j-cluster with $\mathcal{M}, \vec{w}_1 \models \chi$, $\mathcal{M}, \vec{w}_2 \models \chi$, and $V^\star_{\mathbf{C}}(c, \vec{w}_1) \neq V^\star_{\mathbf{C}}(c, \vec{w}_2)$. By IH, $\mathcal{M}_s, \vec{w}_1 \models \chi$ and $\mathcal{M}_s, \vec{w}_2 \models \chi$. Since $\vec{h}$ is in the same cluster as $\vec{p}, \vec{w}_1, \vec{w}_2$, $\mathcal{M}_s, \vec{h} \models \neg Kv_j(\chi, S^k c)$.

Finite Model Property

($\Rightarrow$) Assume $\mathcal{M}_s, \vec{h} \models \neg Kv_j(\chi, S^k c)$. By semantics, there exist $\vec{z}_1, \vec{z}_2 \in W_s$ such that $\vec{h} \sim_j^s \vec{z}_1$, $\vec{h} \sim_j^s \vec{z}_2$, $\mathcal{M}_s, \vec{z}_1 \models \chi$, $\mathcal{M}_s, \vec{z}_2 \models \chi$, and $V_{\mathbf{C}}^s(c, \vec{z}_1) \neq V_{\mathbf{C}}^s(c, \vec{z}_2)$. Since $\vec{z}_1, \vec{z}_2$ are in the same j-cluster as $\vec{h}$, $\mathrm{height}(\vec{z}_\ell) \leq \mathrm{height}(\vec{h}) + 1 \leq D - \mathrm{depth}(\chi)$, the IH gives $\mathcal{M}, \vec{z}_\ell \models \chi$ ($\ell = 1, 2$). Moreover, $V_{\mathbf{C}}^\star(c, \vec{z}_1) = V_{\mathbf{C}}^s(c, \vec{z}_1) \neq V_{\mathbf{C}}^s(c, \vec{z}_2) = V_{\mathbf{C}}^\star(c, \vec{z}_2)$. Since $\vec{h} \sim_j' \vec{z}_1$ and $\vec{h} \sim_j' \vec{z}_2$ in $\mathcal{M}$, we get $\mathcal{M}, \vec{h} \models \neg Kv_j(\chi, S^k c)$.

From Non-standard Models to Standard Models

Lemma

For any formula ϕ satisfiable in a finite non-standard model $\mathcal{N}$, it is also satisfiable in a finite standard model.

From Non-standard Models to Standard Models

Let $\mathcal{N} = \langle W, \mathbb{N}^\star, \mathbf{S}^\star, \{\sim_i\}_{i\in\mathbf{I}}, V^\star_{\mathbf{C}}\rangle$ be a non-standard model such that $\mathcal{N}, w_0 \models \phi$, where W is finite. Let

$$U = \{V^\star_{\mathbf{C}}(c, w) \mid c \text{ occurs in } \phi,\ w \in W\} \cup \{0\}.$$

Since W is finite and ϕ contains only finitely many constants, U is finite and thus $U \cap Z_n$ and $U \cap \mathbb{N}$ are finite for each n.

Let $B = \max(\{a + b \mid \exists x, y \in \mathbf{C} \cup \{0\},\ S^a x \approx S^b y \text{ is a subformula of } \phi\} \cup \{0\})$. We build an injective map $f : U \to \mathbb{N}$ such that:

1. For $x, y \in U$ in the same component (both in $\mathbb{N}$ or both in the same Z_n), $f(x) - f(y) = x - y$.
2. For $x, y \in U$ in different components, $|f(x) - f(y)| > B$.
3. $f(x) = x$ for all $x \in U \cap \mathbb{N}$.

From Non-standard Models to Standard Models

Define a standard model $\mathcal{M} = \langle W, \mathbb{N}, \mathbf{S}, \{\sim_i\}_{i\in\mathbf{I}}, V_{\mathbf{C}}\rangle$ by defining $V_{\mathbf{C}}(c,u) = f\big(V^{\star}_{\mathbf{C}}(c,u)\big)$. We show by induction on subformulas ψ of ϕ that for all $u \in W, \mathcal{N}, u \models \psi \iff \mathcal{M}, u \models \psi$.

- Atomic case: $\psi := S^a c_1 \approx S^b c_2$ (the case of $S^k 0$ can be proved similarly). If $\mathcal{N}, u \models S^a c_1 \approx S^b c_2$, then $V^{\star}_{\mathbf{C}}(c_1,u) + a = V^{\star}_{\mathbf{C}}(c_2,u) + b$. By the definition of f, we have $f(V^{\star}_{\mathbf{C}}(c_1,u)) + a = f(V^{\star}_{\mathbf{C}}(c_2,u)) + b$, which means $\mathcal{M}, u \models S^a c_1 \approx S^b c_2$. Suppose $\mathcal{N}, u \models \neg S^a c_1 \approx S^b c_2$, then if $V^{\star}_{\mathbf{C}}(c_1,u)$ and $V^{\star}_{\mathbf{C}}(c_2,u)$ are in the same component but $V^{\star}_{\mathbf{C}}(c_1,u) + a \neq V^{\star}_{\mathbf{C}}(c_2,u) + b$, then by property 1, $f(V^{\star}_{\mathbf{C}}(c_1,u)) + a \neq f(V^{\star}_{\mathbf{C}}(c_2,u)) + b$ and $\mathcal{M}, u \models \neg S^a c_1 \approx S^b c_2$. If $V^{\star}_{\mathbf{C}}(c_1,u)$ and $V^{\star}_{\mathbf{C}}(c_2,u)$ are in different components, then by property 2, $|f(V^{\star}_{\mathbf{C}}(c_1,u)) - f(V^{\star}_{\mathbf{C}}(c_2,u))| > B \geq |a-b|$, so $f(V^{\star}_{\mathbf{C}}(c_1,u)) + a \neq f(V^{\star}_{\mathbf{C}}(c_2,u)) + b$ and $\mathcal{M}, u \models \neg S^a c_1 \approx S^b c_2$.

From Non-standard Models to Standard Models

- $\psi := K_i\chi$. The IH is: $\forall v \in W, \mathcal{N}, v \models \chi \iff \mathcal{M}, v \models \chi$.

$$\begin{aligned}
\mathcal{N}, u \models K_i\chi &\iff \forall v \in W(u \sim_i v \Rightarrow \mathcal{N}, v \models \chi) \\
&\iff \forall v \in W(u \sim_i v \Rightarrow \mathcal{M}, v \models \chi) \quad \text{(by IH)} \\
&\iff \mathcal{M}, u \models K_i\chi
\end{aligned}$$

- $\psi := Kv_i(\chi, c)$. Note that the IH is: $\forall v \in W, \mathcal{N}, v \models \chi \iff \mathcal{M}, v \models \chi$.

$$\begin{aligned}
\mathcal{N}, u \models Kv_i(\chi, c) &\iff \forall v_1, v_2 \sim_i u, \text{ if } \mathcal{N}, v_1 \models \chi \text{ and } \mathcal{N}, v_2 \models \chi, \\
&\qquad \text{then } V^\star_{\mathbf{C}}(c, v_1) = V^\star_{\mathbf{C}}(c, v_2) \\
&\iff \forall v_1, v_2 \sim_i u, \text{if } \mathcal{M}, v_1 \models \chi \text{ and } \mathcal{M}, v_2 \models \chi, \\
&\qquad \text{then } f(V^\star_{\mathbf{C}}(c, v_1)) = f(V^\star_{\mathbf{C}}(c, v_2)) \quad \text{(IH, } f \text{ is injective)} \\
&\iff \forall v_1, v_2 \sim_i u, \text{if } \mathcal{M}, v_1 \models \chi \text{ and } \mathcal{M}, v_2 \models \chi, \\
&\qquad \text{then } V_{\mathbf{C}}(c, v_1) = V_{\mathbf{C}}(c, v_2) \\
&\iff \mathcal{M}, u \models Kv_i(\chi, c)
\end{aligned}$$

Weak Completeness w.r.t. Standard Models

Theorem

$\mathbb{ELKVSA}^r$ *is weakly complete w.r.t. standard models.*

Corollary

$\mathbf{ELKvSA}^r$ *has finite model property w.r.t. standard models.*

Theorem

$\mathbf{ELKvSA}^r$ *is decidable.*

Proof of Decidability

Given a satisfiable formula ϕ, it is satisfiable in a finite standard model $\mathcal{M} = \langle W, \mathbb{N}, \mathbf{S}, \{\sim_i\}_{i \in I}, V_{\mathbf{C}} \rangle$ of size $f(|\phi|)$, where f is a computable function. Let C_ϕ be the set of constants occurring in ϕ. And let $B = \max(\{a + b \mid \exists x, y \in \mathbf{C} \cup \{0\},\ S^a x \approx S^b y \text{ is a subformula of } \phi\} \cup \{0\})$. Let $U = \{V_{\mathbf{C}}(c, w) \mid c \in C_\phi, w \in W\} \cup \{0\}$. We have $|U| \leq |C_\phi| \cdot f(|\phi|) + 1$. Construct an injective map $g : U \to \mathbb{N}$ as follows: Start from $g(0) = 0$, consider the least number $x \in U$ such that $g(x)$ is not defined. Suppose the greatest number $y \in U$ with $g(y)$ defined is $g(y) = m$. If $x - y \leq B$, let $g(x) = m + (x - y)$; otherwise, let $g(x) = m + B + 1$. Let $\mathcal{M}' = \langle W, \mathbb{N}, \mathbf{S}, \{\sim_i\}_{i \in I}, V'_{\mathbf{C}} \rangle$ be the new model defined by $V'_{\mathbf{C}}(c, w) = g(V_{\mathbf{C}}(c, w))$ for each $c \in C_\phi$ and $w \in W$.

Proof of Decidability

By a similar argument, we can show that $\mathcal{M}', w \models \phi$. Hence if ϕ is satisfiable, it is satisfiable in a finite standard model, where the size of the model is bounded by $f(|\phi|)$ and $V_{\mathbf{C}}(c, w) \in \{0, 1, \cdots, (|C_\phi| \cdot f(|\phi|) + 1) \cdot (B + 1)\}$ for each $c \in C_\phi$ and $w \in W$. Then we can enumerate all such models (the size of the model is bounded, the codomain of $V_{\mathbf{C}}$ is bounded, and we only need to consider agents occurring in ϕ and constants in C_ϕ) and check if ϕ is satisfiable in any of them. There are only finitely many such models, and checking satisfiability in each model is decidable, so we can decide the satisfiability of ϕ. Hence $\mathbf{ELKvSA}^r$ is decidable.

Back to the Puzzle

Definition

Let the set of premises be $\Gamma_0 = \{Kv_A a, Kv_B b, \neg a \approx 0 \rightarrow \neg Kv_A b, \neg b \approx 0 \rightarrow \neg Kv_B a\} \cup \{K_i(a \approx Sb \vee b \approx Sa) \mid i \in \{A, B, C\}\}$. And $\Gamma_{n+1} := \Gamma_n \cup \{K_A\psi, K_B\psi, K_C\psi \mid \psi \in \Gamma_n\}$. The conclusion is:

$$[\neg Kv_A b][\neg Kv_B a][Kv_A b](Kv_B a \wedge K_C((a \approx SS0 \wedge b \approx SSS0) \vee (b \approx SS0 \wedge a \approx S0)))$$

The premise $\neg b \approx 0 \rightarrow \neg Kv_B a$ warrants some explanation. Consider the case $K_A(b \approx 5 \rightarrow Kv_B a)$.

Back to the Puzzle

For convenience, we use the following notations:
$\chi_m^n := \wedge_0^m \neg a \approx S^k 0 \wedge \wedge_0^n \neg b \approx S^k 0$, $\varphi_1 := \neg K v_A b$. And

$$\varphi_n := \begin{cases} \varphi_{n-1} \wedge [\varphi_{n-1}] \neg K v_B a & \text{if } n \text{ is even} \\ \varphi_{n-1} \wedge [\varphi_{n-1}] \neg K v_A b & \text{if } n \text{ is odd} \end{cases}$$

$$\psi_n := \begin{cases} [\varphi_{n-1}] K v_B a & \text{if } n \text{ is even} \\ [\varphi_{n-1}] K v_A b & \text{if } n \text{ is odd} \end{cases}$$

Back to the Puzzle

Proposition

1. *If n is even,* $\Gamma_{n-1} \vdash (\varphi_n \leftrightarrow \chi_{n-2}^{n-1}) \wedge (\psi_n \leftrightarrow \neg\chi_{n-2}^{n-1})$
2. *If n is odd,* $\Gamma_{n-1} \vdash (\varphi_n \leftrightarrow \chi_{n-1}^{n-2}) \wedge (\psi_n \leftrightarrow \neg\chi_{n-1}^{n-2})$.

Proposition

1. *If n is even,* $\Gamma_n \vdash \underbrace{[\neg Kv_A b][\neg Kv_B a] \cdots [\neg Kv_B a]}_{n}[Kv_A b](Kv_B a \wedge K_C((a \approx S^n 0 \wedge b \approx S^{n+1}0) \vee (b \approx S^n 0 \wedge a \approx S^{n-1}0)))$.
2. *If n is odd,* $\Gamma_n \vdash \underbrace{[\neg Kv_A b][\neg Kv_B a] \cdots [\neg Kv_A b]}_{n}[Kv_B a](Kv_A b \wedge K_C((b \approx S^n 0 \wedge a \approx S^{n+1}0) \vee (a \approx S^n 0 \wedge b \approx S^{n-1}0)))$.

Conclusion and Future Work

Where My Ideas Come From

- The whole idea of the paper: Prof. Wang.
- The idea of using rules: Bo Hong, 2023.
- The idea of valuation game: Wang and Fan, 2013.
- The idea of tree-unraveling: similar to Xiong, 2014.
- The idea of using non-standard models: inspired by Prof. Ding.
- No fake formula: too difficult.
- No filtration and no Henkin set: failure of many attempts.

Generalize to $(\mathbb{N}, +, \times, 0, 1)$: Impossible

Hilbert's tenth problem: there is no algorithm to determine whether a given polynomial equation with integer coefficients has an integer (natural number) solution.

If we have a recursive axiom system, then we can enumerate all the polynomial equations without natural number solutions, and since the set of polynomial equations with integer solutions is recursively enumerable, we can decide whether a given polynomial equation has a natural number solution

Other Future Work

- Consider other frame classes except S5.
- Is $\mathbf{ELKvSA}^r$ in PSPACE?
- Solve Bo Hong's *KvP*.
- Generalize to any language and any theory *T*.
- Consider non-rigid function symbols. Consider varying domain.
- Combining mathematical theories with philosophical theories.

Thanks!